

Final Plan: Fine-Tune Plan 1 model on ShanghaiTech A+B

Strategy: Plan 1 Model -> fine-tune on Part A+B+C(India) combined (70 images) -> adaptive sigma -> training

Cell 0: Mount Drive

```
from google.colab import drive
drive.mount('/content/drive')
import os
BASE='/content/drive/MyDrive/DLENDSEM'
PLAN1=f'{BASE}/plan1_csrnet'
FINAL=f'{BASE}/final_v2'
SHA=f'{BASE}/dataset/ShanghaiTech'
for d in
[f'{FINAL}/density_maps_train_B',f'{FINAL}/density_maps_test_B',f'{FINAL}/checkpoints',f'{FINAL}/results',f'{FINAL}/visualizations']:
    os.makedirs(d,exist_ok=True)
print('Folders ready')
```

Mounted at /content/drive
Folders ready

```
DATASET_DIR = '/content/drive/MyDrive/DLENDSEM/dataset'
SHA_DIR = f'{DATASET_DIR}' # Corrected: ShanghaiTech content is directly in DATASET_DIR
```

```
import os
import glob # Import glob here

if os.path.exists(f'{SHA_DIR}/part_A_final/train_data/images') and len(glob.glob(f'{SHA_DIR}/part_A_final/train_data/images/*.jpg')) > 0:
    print("Dataset already exists in Drive and is complete, skipping download.")
else:
    !pip install gdown -q
    !gdown 16dhJn7k4FWVwByRsQAEpl9lwjuV03jVI -O /content/shanghaitech.zip
    # Use -o to force overwrite without prompting
    !unzip -o -q /content/shanghaitech.zip -d {DATASET_DIR}
    print("Dataset downloaded and extracted to Drive.")
```

Dataset already exists in Drive and is complete, skipping download.

```
!pip install bing-image-downloader -q
from bing_image_downloader import downloader
```

```

import shutil, glob, os

FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
C_DIR =
'/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech/part_C_india'
for d in [f'{C_DIR}/train_data/images', f'{C_DIR}/test_data/images',
          f'{FINAL}/density_maps_train_C',
          f'{FINAL}/density_maps_test_C']:
    os.makedirs(d, exist_ok=True)

# Download ~100 images across multiple queries
queries = [
    'indian crowd gathering aerial view',
    'kumbh mela crowd',
    'indian railway station crowd',
    'indian festival crowd dense',
    'indian market crowd top view',
    'india election rally crowd',
    'indian temple crowd gathering',
    'indian street crowd busy',
]
tmp = '/content/indian_crowd_tmp'
for q in queries:
    try:
        downloader.download(q, limit=15, output_dir=tmp,
                             adult_filter_off=False,
                             force_replace=False, timeout=30)
    except: pass

# Collect all downloaded images
all_imgs = []
for ext in ['*.jpg', '*.jpeg', '*.png', '*.bmp']:
    all_imgs.extend(glob.glob(f'{tmp}/**/{ext}', recursive=True))
print(f'Downloaded: {len(all_imgs)} images')

# Filter valid images and rename sequentially
from PIL import Image
valid = []
for ip in all_imgs:
    try:
        img = Image.open(ip).convert('RGB')
        w, h = img.size
        if w >= 256 and h >= 256: # skip tiny images
            valid.append(ip)
    except: pass
print(f'Valid (>=256px): {len(valid)}')

# Split: first 70 train, next 15 test
import random
random.seed(42)

```

```

random.shuffle(valid)
train_imgs = valid[:70]
test_imgs = valid[70:85]

for i, ip in enumerate(train_imgs):
    img = Image.open(ip).convert('RGB')
    img.save(f'{C_DIR}/train_data/images/IMG_C_{i+1:03d}.jpg')
for i, ip in enumerate(test_imgs):
    img = Image.open(ip).convert('RGB')
    img.save(f'{C_DIR}/test_data/images/IMG_C_{i+1:03d}.jpg')

print(f'Part C: {len(train_imgs)} train, {len(test_imgs)} test')
print(f'Saved to: {C_DIR}')

```

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https://static.vecteezy.com/system/resources/previews/024/250/121/large_2x/ai-generated-people-crowd-gathering-in-the-street-aerial-view-photo.jpg

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[%] Downloading Image #11 from <https://c8.alamy.com/comp/2EMRE8B/kolkata-india-april-3-crowd-during-prime-minister-narendra-modis-political-rally-at-brigade-parade-ground-ahead-of-loksabha-election-on-april-3-2019-in-kolkata-india-photo-by-samir-janahindustan-timessipa-usa-2EMRE8B.jpg>

[%] File Downloaded !

[!!!]Indexing page: 2

[%] Indexed 35 Images on Page 2.

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[%] Downloading Image #12 from <https://thumbs.dreamstime.com/b/crowd-election-rally-india-348659620.jpg>

[%] File Downloaded !

[%] Downloading Image #13 from
https://www.shutterstock.com/shutterstock/photos/198507749/display_1500/stock-photo-vadodara-gujarat-india-april-large-crowd-walking-with-gujarat-chief-minister-and-bjp-198507749.jpg
[%] File Downloaded !

[%] Downloading Image #14 from <https://www.shutterstock.com/image-photo/priyanka-gandhi-election-campaign-crowd-600w-2123836211.jpg>
[%] File Downloaded !

[%] Downloading Image #15 from
https://www.theindiaforum.in/sites/default/files/field/image/2025/IMG_8010.JPG
[%] File Downloaded !

[%] Done. Downloaded 15 images.
[%] Downloading Images to /content/indian_crowd_tmp/indian temple crowd gathering

[!!]Indexing page: 1

[%] Indexed 15 Images on Page 1.

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[%] Downloading Image #1 from
<https://thumbs.dreamstime.com/b/pilgrims-celebrating-festival-around-temple-india-crowd-gathering-around-temple-festival-celebration-india-ai-377167932.jpg>
[%] File Downloaded !

[%] Downloading Image #2 from
<https://images.pexels.com/photos/19553179/pexels-photo-19553179/free-photo-of-crowd-at-hindu-temple.jpeg?auto=compress&cs=tinysrgb&w=1260&h=750&dpr=1>
[%] File Downloaded !

[%] Downloading Image #3 from
<https://images.pexels.com/photos/19553179/pexels-photo-19553179.jpeg?auto=compress&cs=tinysrgb&h=627&fit=crop&w=1200>
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[%] Downloading Image #4 from https://images.pond5.com/hindu-gathering-crowd-murugan-temple-footage-235628300_iconl.jpeg
[%] File Downloaded !

[%] Downloading Image #5 from

<https://images.playground.com/c2675a9e2bc641d986318e7e0395d31c.jpeg>

[!] Issue getting:

<https://images.playground.com/c2675a9e2bc641d986318e7e0395d31c.jpeg>

[!] Error:: HTTP Error 404: Not Found

[%] Downloading Image #5 from <https://thumbs.dreamstime.com/b/indian-traditional-public-crowd-temple-navratri-festival-india-officer-duty-department-staff-night-lord-ravana-rally-163110057.jpg?w=1600>

[%] File Downloaded !

[%] Downloading Image #6 from

<https://images.pexels.com/photos/18365480/pexels-photo-18365480.jpeg?cs=srgb&dl=pexels-kolkatarchobiwala-18365480.jpg&fm=jpg>

[%] File Downloaded !

[%] Downloading Image #7 from

<https://images.pexels.com/photos/12387055/pexels-photo-12387055.jpeg?cs=srgb&dl=pexels-subhajit-paul-12387055.jpg&fm=jpg>

[%] File Downloaded !

[%] Downloading Image #8 from

https://images.stockcake.com/public/c/a/2/ca2b8323-528d-4b78-9532-df26169bab67_large/temple-festive-crowd-stockcake.jpg

[%] File Downloaded !

[%] Downloading Image #9 from <https://thumbs.dreamstime.com/z/indian-temple-mass-number-crowd-peoples-standing-front-temple-image-has-taken-indian-street-202977035.jpg>

[%] File Downloaded !

[%] Downloading Image #10 from <https://thumbs.dreamstime.com/z/inside-indian-temple-delhi-gurdwara-sis-ganj-sahib-indian-temple-old-delhi-india-137588819.jpg>

[%] File Downloaded !

[%] Downloading Image #11 from

<https://images.pexels.com/photos/16890524/pexels-photo-16890524.jpeg?cs=srgb&dl=pexels-tkirkgoz-16890524.jpg&fm=jpg>

[%] File Downloaded !

[%] Downloading Image #12 from

https://img.freepik.com/premium-photo/painting-depicting-large-gathering-indian-temple_1282444-190386.jpg

[%] File Downloaded !

[%] Downloading Image #13 from <https://c8.alamy.com/comp/2fgf86d/busy-crowd-at-temple-on-ghat-maheshwar-india-2fgf86d.jpg>

[%] File Downloaded !

[%] Downloading Image #14 from <https://thumbs.dreamstime.com/b/crowd-indian-temple-opening-ceremony-15993231.jpg>

[%] File Downloaded !

[!!]Indexing page: 2

[%] Indexed 35 Images on Page 2.

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[%] Downloading Image #15 from
https://img.freepik.com/premium-photo/indian-temple-with-people-crowd_394462-2367.jpg
[%] File Downloaded !

[%] Done. Downloaded 15 images.
[%] Downloading Images to /content/indian_crowd_tmp/indian street crowd busy

[!!]Indexing page: 1

[%] Indexed 15 Images on Page 1.

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[%] Downloading Image #1 from
<https://c8.alamy.com/comp/2ST7A8P/street-scene-indian-people-crowd-setting-sun-on-a-busy-mumbai-bombay-street-maharashtra-india-asia-2ST7A8P.jpg>
[%] File Downloaded !

[%] Downloading Image #2 from
<https://videocdn.cdnpk.net/videos/1c33f7be-c082-441e-9381-65ae4cdd4333/horizontal/thumbnails/large.jpg>
[Error]Invalid image, not saving
<https://videocdn.cdnpk.net/videos/1c33f7be-c082-441e-9381-65ae4cdd4333/horizontal/thumbnails/large.jpg>

[!] Issue getting: <https://videocdn.cdnpk.net/videos/1c33f7be-c082-441e-9381-65ae4cdd4333/horizontal/thumbnails/large.jpg>
[!] Error:: Invalid image, not saving
<https://videocdn.cdnpk.net/videos/1c33f7be-c082-441e-9381-65ae4cdd4333/horizontal/thumbnails/large.jpg>

[%] Downloading Image #2 from
https://as1.ftcdn.net/jpg/10/79/11/08/1000_F_1079110864_9WJ4PJs4Pn0d7GjqvTMCr3WkLzJAE535.jpg
[%] File Downloaded !

[%] Downloading Image #3 from
https://st4.depositphotos.com/1006214/22040/i/450/depositphotos_220402862-stock-photo-new-delhi-india-october-2015.jpg
[%] File Downloaded !

[%] Downloading Image #4 from
<https://images.pexels.com/photos/14994013/pexels-photo-14994013.jpeg?cs=srgb&dl=pexels-deepak-ghoran-14994013.jpg&fm=jpg>
[%] File Downloaded !

[%] Downloading Image #5 from <https://c8.alamy.com/comp/DPD08T/a-typically-busy-indian-street-DPD08T.jpg>
[%] File Downloaded !

[%] Downloading Image #6 from https://images.pond5.com/busy-pedestrian-street-indian-city-footage-249250785_icon1.jpeg
[%] File Downloaded !

[%] Downloading Image #7 from
<https://images.pexels.com/photos/32592264/pexels-photo-32592264/free-photo-of-bustling-indian-street-scene-with-diverse-crowd.jpeg?auto=compress&cs=tinysrgb&w=1260&h=750&dpr=2>
[%] File Downloaded !

[%] Downloading Image #8 from
<https://c8.alamy.com/comp/WTF56H/crowded-busy-indian-street-WTF56H.jpg>
[%] File Downloaded !

[%] Downloading Image #9 from <https://c8.alamy.com/comp/ETHGMC/busy-indian-street-scene-jaisalmer-rajasthan-india-ETHGMC.jpg>
[%] File Downloaded !

[%] Downloading Image #10 from <https://c8.alamy.com/comp/C3A4C6/fast-moving-traffic-on-a-busy-indian-street-C3A4C6.jpg>
[%] File Downloaded !

[%] Downloading Image #11 from <https://c8.alamy.com/comp/DPD09E/a-typically-busy-indian-street-DPD09E.jpg>
[%] File Downloaded !

[%] Downloading Image #12 from <https://c8.alamy.com/comp/C3A4K6/fast-moving-traffic-on-a-busy-indian-street-C3A4K6.jpg>
[%] File Downloaded !

[%] Downloading Image #13 from <https://www.shutterstock.com/image-photo/delhi-india-january-16-people-600w-325628909.jpg>
[%] File Downloaded !

[%] Downloading Image #14 from <https://c8.alamy.com/comp/3C40YGC/busy->

```
street-with-dense-crowd-and-vehicles-in-an-indian-market-varanasi-uttar-pradesh-india-3C40YGC.jpg
[%] File Downloaded !
```

```
[!!!]Indexing page: 2
```

```
[%] Indexed 35 Images on Page 2.
```

```
=====
```

```
[%] Downloading Image #15 from https://thumbs.dreamstime.com/b/cycle-rickshaw-drive-busy-asian-street-full-pedestrians-jaipur-india-jaipur-population-people-capital-51188176.jpg
[%] File Downloaded !
```

```
[%] Done. Downloaded 15 images.
Downloaded: 119 images
Valid (>=256px): 118
Part C: 70 train, 15 test
Saved to:
/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech/part_C_india
```

Cell 1: Install & Imports

```
!pip install scipy h5py matplotlib opencv-python-headless gdown -q
import torch,torch.nn as nn,torch.optim as optim
from torch.utils.data import Dataset,DataLoader
from torchvision import transforms,models
from PIL import Image
import numpy as np,h5py,cv2,glob,os,random,json,time
import scipy.io as io
from tqdm import tqdm
from scipy.ndimage import gaussian_filter
from scipy.spatial import KDTree
print(f'PyTorch {torch.__version__} | GPU:
{torch.cuda.get_device_name(0)}')
```

```
PyTorch 2.10.0+cu128 | GPU: Tesla T4
```

Cell 2: Download Plan 1 CSRNet Pretrained Weights

From the first plan trained on ShanghaiA model with MAE: 76.25 | MSE: 116.00.

```
import gdown
import os
```

```

FINAL='/content/drive/MyDrive/DLENDSEM/final_plan'
PRETRAINED=f'{FINAL}/pretrained/plan_1_partA.pth.tar'

# Create the directory if it doesn't exist
os.makedirs(os.path.dirname(PRETRAINED), exist_ok=True)

if not os.path.exists(PRETRAINED):
    print('Downloading PLSN 1 CSRNet Part A weights (MAE: 76.25 | MSE: 116.00)...')
    gdown.download(id='1Z-atzS5Y2p0d-nEwqZRVBDMYJDreGWHH', output=PRETRAINED, quiet=False)
    print('Downloaded!')
else:
    print(f'Already have: {PRETRAINED}')

Downloading PLSN 1 CSRNet Part A weights (MAE: 76.25 | MSE: 116.00)...

Downloading...
From (original): https://drive.google.com/uc?id=1Z-atzS5Y2p0d-nEwqZRVBDMYJDreGWHH
From (redirected): https://drive.google.com/uc?id=1Z-atzS5Y2p0d-nEwqZRVBDMYJDreGWHH&confirm=t&uuid=27980e93-740f-46ae-a9fd-b1600ebbe125
To:
/content/drive/MyDrive/DLENDSEM/final_plan/pretrained/plan_1_partA.pth.tar
100%|██████████| 130M/130M [00:03<00:00, 38.7MB/s]

Downloaded!

```

Cell 3: Adaptive Sigma Density Maps (A+B)

```

import os
import glob

# Inspect a sample ground-truth directory to check file naming convention
SHA='/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech'
# Corrected path: often 'ground-truth' instead of 'ground_truth'
sample_gt_dir = f'{SHA}/part_A/train_data/ground_truth' # Changed hyphen to underscore
print(f'Listing contents of: {sample_gt_dir}')
# List the first 10 files to get an idea of the naming
try:
    files = sorted(os.listdir(sample_gt_dir))
    for i, f in enumerate(files):
        if i < 10:

```

```

        print(f)
    else:
        break
    if not files:
        print(f'No files found in {sample_gt_dir}. This might indicate
an empty directory or incorrect path.')
except FileNotFoundError:
    print(f"Error: Directory '{sample_gt_dir}' not found. Please
ensure the ShanghaiTech dataset is correctly uploaded and the path is
accurate.")
    print("Common variations: 'ground-truth' vs 'ground_truth'. This
code assumes 'ground_truth'.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")

```

```

# Also check if any .mat files exist without the GT_ prefix
print('\nChecking for .mat files without "GT_" prefix:')
try:
    non_gt_files = glob.glob(f'{sample_gt_dir}/*.mat')
    if non_gt_files:
        print(f'Found {len(non_gt_files)} .mat files. Sample:
{os.path.basename(non_gt_files[0])}')
    else:
        print('No .mat files found in the ground-truth directory.')
except FileNotFoundError:
    # This might happen if sample_gt_dir was not found above and glob
is called on a non-existent path
    print(f"Cannot check for .mat files as directory '{sample_gt_dir}'
does not exist.")

```

```

Listing contents of:
/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech/part_A/train_data
/ground_truth
GT_IMG_1.mat
GT_IMG_10.mat
GT_IMG_100.mat
GT_IMG_101.mat
GT_IMG_102.mat
GT_IMG_103.mat
GT_IMG_104.mat
GT_IMG_105.mat
GT_IMG_106.mat
GT_IMG_107.mat

```

```

Checking for .mat files without "GT_" prefix:
Found 300 .mat files. Sample: GT_IMG_110.mat

```

```

import os
import glob

```

```

from PIL import Image
import numpy as np
import h5py
from scipy.ndimage import gaussian_filter
import scipy.io as io
from tqdm import tqdm

def gen_density_fixed(img_path, gt_path, sigma=15):
    img = Image.open(img_path)
    iw, ih = img.size
    mat = io.loadmat(gt_path)
    pts = mat['image_info'][0,0][0,0][0]
    density = np.zeros((ih, iw), dtype=np.float32)
    if len(pts) == 0:
        return density
    for p in pts:
        x = min(int(round(p[0])), iw-1)
        y = min(int(round(p[1])), ih-1)
        density[y, x] = 1.0
    density = gaussian_filter(density, sigma=sigma)
    return density

def process_split(img_dir, gt_dir, out_dir, label):
    os.makedirs(out_dir, exist_ok=True) # Add this line to create the
output directory
    imgs = sorted(glob.glob(f'{img_dir}/*.jpg'))
    done = 0
    for ip in tqdm(imgs, desc=f'[{label}] Processing images'):
        fname = os.path.basename(ip).replace('.jpg', '')
        h5p = f'{out_dir}/{fname}.h5'
        if os.path.exists(h5p):
            done += 1; continue
        gp = f'{gt_dir}/GT_{fname}.mat'
        if not os.path.exists(gp): continue
        den = gen_density_fixed(ip, gp)
        with h5py.File(h5p, 'w') as hf:
            hf['density'] = den
        done += 1
    print(f'[{label}] {done}/{len(imgs)} done')

SHA = '/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech'
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
process_split(f'{SHA}/part_A/train_data/images',
f'{SHA}/part_A/train_data/ground_truth',
f'{FINAL}/density_maps_train_A', 'A Train')
process_split(f'{SHA}/part_A/test_data/images',
f'{SHA}/part_A/test_data/ground_truth',
f'{FINAL}/density_maps_test_A', 'A Test')
process_split(f'{SHA}/part_B/train_data/images',
f'{SHA}/part_B/train_data/ground_truth',

```

```

f'{FINAL}/density_maps_train_B', 'B Train')
process_split(f'{SHA}/part_B/test_data/images',
f'{SHA}/part_B/test_data/ground_truth',
f'{FINAL}/density_maps_test_B', 'B Test')
print('Density maps done!')

[A Train] Processing images: 100%|██████████| 300/300 [00:43<00:00,
6.88it/s]

[A Train] 300/300 done

[A Test] Processing images: 100%|██████████| 182/182 [02:36<00:00,
1.16it/s]

[A Test] 182/182 done

[B Train] Processing images: 100%|██████████| 400/400 [05:12<00:00,
1.28it/s]

[B Train] 400/400 done

[B Test] Processing images: 100%|██████████| 316/316 [04:02<00:00,
1.30it/s]

[B Test] 316/316 done
Density maps done!

```

Cell 4: Dataset & Model

```

class CrowdDataset(Dataset):
    def __init__(self, img_paths, h5_dirs, crop=256, augment=True):
        self.img_paths = img_paths
        self.h5_dirs = h5_dirs if isinstance(h5_dirs, list) else
[h5_dirs]
        self.crop = crop
        self.augment = augment
        self.tf = transforms.Compose([
            transforms.ToTensor(),
            transforms.Normalize([0.485,0.456,0.406],
[0.229,0.224,0.225])
        ])
    def __len__(self): return len(self.img_paths)
    def _find_h5(self, fname):
        for d in self.h5_dirs:
            p = os.path.join(d, f'{fname}.h5')
            if os.path.exists(p): return p
        return None
    def __getitem__(self, idx):
        img = Image.open(self.img_paths[idx]).convert('RGB')

```



```

        fname =
os.path.basename(self.img_paths[idx]).replace('.jpg','')
        with h5py.File(self._find_h5(fname), 'r') as f:
            den = np.array(f['density'], dtype=np.float32)
        w, h = img.size
        cs = self.crop
        # Pad if smaller than crop
        if h < cs or w < cs:
            pad_h, pad_w = max(cs-h, 0), max(cs-w, 0)
            img = np.array(img)
            img = np.pad(img, ((0,pad_h),(0,pad_w),(0,0)),
mode='reflect')
            img = Image.fromarray(img)
            den = np.pad(den, ((0,pad_h),(0,pad_w)), mode='constant')
            w, h = img.size
        # Random crop
        if self.augment:
            t = random.randint(0, h - cs)
            l = random.randint(0, w - cs)
        else:
            t, l = (h-cs)//2, (w-cs)//2
        img = img.crop((l, t, l+cs, t+cs))
        den = den[t:t+cs, l:l+cs]
        # Random horizontal flip
        if self.augment and random.random() > 0.5:
            img = img.transpose(Image.FLIP_LEFT_RIGHT)
            den = np.fliplr(den).copy()
        # Downsample density to 1/8
        den = cv2.resize(den, (cs//8, cs//8),
interpolation=cv2.INTER_CUBIC) * 64
        return self.tf(img), torch.tensor(den).unsqueeze(0)

class CSRNet(nn.Module):
    def __init__(self):
        super().__init__()
        vgg = models.vgg16(weights=models.VGG16_Weights.IMAGENET1K_V1)
        self.frontend = nn.Sequential(*list(vgg.features.children())
[:23])
        self.backend = nn.Sequential(
            nn.Conv2d(512,512,3,padding=2,dilation=2), nn.ReLU(True),
            nn.Conv2d(512,512,3,padding=2,dilation=2), nn.ReLU(True),
            nn.Conv2d(512,512,3,padding=2,dilation=2), nn.ReLU(True),
            nn.Conv2d(512,256,3,padding=2,dilation=2), nn.ReLU(True),
            nn.Conv2d(256,128,3,padding=2,dilation=2), nn.ReLU(True),
            nn.Conv2d(128,64, 3,padding=2,dilation=2), nn.ReLU(True))
        self.output_layer = nn.Conv2d(64, 1, 1)
        for m in self.backend.modules():
            if isinstance(m, nn.Conv2d):
                nn.init.normal_(m.weight, std=0.01)

```

```

        if m.bias is not None: nn.init.constant_(m.bias, 0)
    def forward(self, x):
        return self.output_layer(self.backend(self.frontend(x)))

print('Model & Dataset defined')

Model & Dataset defined

```

Cell 5: Build Combined A+B DataLoader

```

PLAN1 = '/content/drive/MyDrive/DLENDSEM/plan1_csrnet'
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
SHA = '/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech'
C_DIR =
'/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech/part_C_india'

imgs_A = sorted(glob.glob(f'{SHA}/part_A/train_data/images/*.jpg'))
imgs_B = sorted(glob.glob(f'{SHA}/part_B/train_data/images/*.jpg'))
imgs_C = sorted(glob.glob(f'{C_DIR}/train_data/images/*.jpg'))
all_imgs = imgs_A + imgs_B + imgs_C

# Part A uses Plan 1 density maps, Part B and C use our new ones
h5_dirs = [f'{PLAN1}/density_maps_train',
f'{FINAL}/density_maps_train_B', f'{FINAL}/density_maps_train_C']

train_ds = CrowdDataset(all_imgs, h5_dirs, crop=256, augment=True)
train_loader = DataLoader(train_ds, batch_size=4, shuffle=True,
num_workers=2, pin_memory=True)
print(f'Combined: {len(train_ds)} images (A:{len(imgs_A)} + B:
{len(imgs_B)} + C:{len(imgs_C)})')

Combined: 770 images (A:300 + B:400 + C:70)

```

Cell 6: Load Plan 1 Weights

```

model=CSRNet().cuda()
FINAL='/content/drive/MyDrive/DLENDSEM/final_plan'
PRETRAINED=f'{PLAN1}/pretrained/plan1_partA.pth.tar'

if isinstance(checkpoint,dict) and 'state_dict' in checkpoint:
    state=checkpoint['state_dict']
    epoch_info=checkpoint.get('epoch','?')
    best_mae=checkpoint.get('best_prec1','?')
    print(f'Best checkpoint: epoch={epoch_info}, best_mae={best_mae}')
else:
    state=checkpoint

new_state={}
for k,v in state.items():
    nk=k.replace('module.','')

```

```

        new_state[nk]=v

missing,unexpected=model.load_state_dict(new_state,strict=False)
if missing: print(f'Missing keys: {missing}')
if unexpected: print(f'Unexpected keys: {unexpected}')
if not missing and not unexpected: print('All weights loaded
perfectly!')
print(f'Total params: {sum(p.numel() for p in model.parameters()):,}')

Best checkpoint: epoch=436, best_mae=76.25
All weights loaded perfectly!
Total params: 16,263,489

```

Cell 7: Training

Early stopping patience=30

```

import torch,torch.nn as nn

NUM_EPOCHS = 500
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'

if 'model' not in locals() or not isinstance(model, nn.Module):

PRETRAINED_LOAD_PATH='/content/drive/MyDrive/DLENDSEM/final_plan/pretr
ained/plan_1_partA.pth.tar'
    model = CSRNet().cuda()
    checkpoint = torch.load(PRETRAINED_LOAD_PATH, map_location='cpu',
weights_only=False)

    if isinstance(checkpoint, dict) and 'state_dict' in checkpoint:
        state = checkpoint['state_dict']
    else:
        state = checkpoint

    new_state = {}
    for k, v in state.items():
        nk = k.replace('module.', '')
        new_state[nk] = v

    model.load_state_dict(new_state, strict=False)
    print('Model re-initialized and pretrained weights loaded for
current cell execution.')

for p in model.frontend.parameters():
    p.requires_grad = False
trainable = sum(p.numel() for p in model.parameters() if
p.requires_grad)
print(f'Trainable: {trainable:,} (backend+output)')

```

```

criterion = nn.MSELoss()
optimizer = torch.optim.Adam(filter(lambda p: p.requires_grad,
model.parameters()), lr=1e-5)
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer,
T_max=NUM_EPOCHS, eta_min=1e-7)

losses = []
existing = sorted(glob.glob(f'{FINAL}/checkpoints/epoch_*.pth'))
start = 0
if existing:
    try:
        # Safely extract epoch number
        epoch_str = existing[-1].split('_epoch_')[1].split('.pth')[0]
        start = int(epoch_str)
        model.load_state_dict(torch.load(existing[-1],
map_location='cpu'))
        model.cuda()
        print(f'Resuming from epoch {start}')
    except (IndexError, ValueError) as e:
        print(f'Warning: Could not parse epoch from {existing[-1]} due
to {e}. Starting from epoch 0.')
        start = 0
lf = f'{FINAL}/results/losses.json'
if os.path.exists(lf):
    with open(lf) as f: losses = json.load(f)

best_loss = float('inf')
no_improve = 0
PATIENCE = 30

t0 = time.time()
for ep in range(start, NUM_EPOCHS):
    model.train()
    ep_loss = 0
    for imgs, dens in train_loader:
        out = model(imgs.cuda())
        loss = criterion(out, dens.cuda())
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()
        ep_loss += loss.item()
    scheduler.step()
    avg = ep_loss / len(train_loader)
    losses.append(avg)
    if avg < best_loss:
        best_loss = avg
        no_improve = 0
        torch.save(model.state_dict(),
f'{FINAL}/checkpoints/best_model.pth')
    else:

```

```

        no_improve += 1
    if (ep+1) % 1 == 0:
        mins = (time.time()-t0)/60
        print(f'[{ep+1}/{NUM_EPOCHS}] Loss:{avg:.6f} | {mins:.1f}min')
    if (ep+1) % 25 == 0:
        torch.save(model.state_dict(),
f'{FINAL}/checkpoints/epoch_{ep+1}.pth')
        with open(lf, 'w') as f: json.dump(losses, f)
    if no_improve >= PATIENCE:
        print(f'Early stopping at epoch {ep+1}')
        break

torch.save(model.state_dict(), f'{FINAL}/checkpoints/final_model.pth')
with open(lf, 'w') as f: json.dump(losses, f)
print('Training complete!')

```

Trainable: 8,628,225 (backend+output)

```

[1/500] Loss:0.001347 | 7.0min
[2/500] Loss:0.001350 | 7.3min
[3/500] Loss:0.001299 | 7.6min
[4/500] Loss:0.001354 | 7.9min
[5/500] Loss:0.001333 | 8.2min
[6/500] Loss:0.001299 | 8.5min
[7/500] Loss:0.001336 | 8.8min
[8/500] Loss:0.001377 | 9.1min
[9/500] Loss:0.001314 | 9.4min
[10/500] Loss:0.001404 | 9.7min
[11/500] Loss:0.001329 | 10.0min
[12/500] Loss:0.001267 | 10.3min
[13/500] Loss:0.001264 | 10.6min
[14/500] Loss:0.001323 | 10.9min
[15/500] Loss:0.001341 | 11.2min
[16/500] Loss:0.001312 | 11.5min
[17/500] Loss:0.001359 | 11.8min
[18/500] Loss:0.001293 | 12.1min
[19/500] Loss:0.001261 | 12.4min
[20/500] Loss:0.001240 | 12.7min
[21/500] Loss:0.001373 | 13.0min
[22/500] Loss:0.001261 | 13.3min
[23/500] Loss:0.001331 | 13.6min
[24/500] Loss:0.001338 | 13.9min
[25/500] Loss:0.001370 | 14.2min
[26/500] Loss:0.001376 | 14.5min
[27/500] Loss:0.001336 | 14.8min
[28/500] Loss:0.001316 | 15.1min
[29/500] Loss:0.001264 | 15.4min
[30/500] Loss:0.001311 | 15.7min
[31/500] Loss:0.001227 | 16.0min
[32/500] Loss:0.001364 | 16.3min
[33/500] Loss:0.001276 | 16.6min

```

[34/500]	Loss:0.001245	16.9min
[35/500]	Loss:0.001308	17.2min
[36/500]	Loss:0.001342	17.5min
[37/500]	Loss:0.001307	17.8min
[38/500]	Loss:0.001214	18.1min
[39/500]	Loss:0.001285	18.4min
[40/500]	Loss:0.001231	18.7min
[41/500]	Loss:0.001344	19.0min
[42/500]	Loss:0.001295	19.3min
[43/500]	Loss:0.001217	19.6min
[44/500]	Loss:0.001289	19.9min
[45/500]	Loss:0.001186	20.2min
[46/500]	Loss:0.001263	20.5min
[47/500]	Loss:0.001227	20.8min
[48/500]	Loss:0.001245	21.1min
[49/500]	Loss:0.001184	21.4min
[50/500]	Loss:0.001193	21.7min
[51/500]	Loss:0.001248	22.0min
[52/500]	Loss:0.001276	22.3min
[53/500]	Loss:0.001224	22.6min
[54/500]	Loss:0.001208	22.9min
[55/500]	Loss:0.001180	23.2min
[56/500]	Loss:0.001166	23.5min
[57/500]	Loss:0.001162	23.8min
[58/500]	Loss:0.001170	24.1min
[59/500]	Loss:0.001212	24.4min
[60/500]	Loss:0.001151	24.7min
[61/500]	Loss:0.001296	25.0min
[62/500]	Loss:0.001267	25.3min
[63/500]	Loss:0.001259	25.6min
[64/500]	Loss:0.001235	25.9min
[65/500]	Loss:0.001176	26.2min
[66/500]	Loss:0.001179	26.5min
[67/500]	Loss:0.001228	26.8min
[68/500]	Loss:0.001232	27.1min
[69/500]	Loss:0.001240	27.4min
[70/500]	Loss:0.001144	27.7min
[71/500]	Loss:0.001283	28.0min
[72/500]	Loss:0.001192	28.3min
[73/500]	Loss:0.001210	28.6min
[74/500]	Loss:0.001222	28.9min
[75/500]	Loss:0.001225	29.2min
[76/500]	Loss:0.001232	29.5min
[77/500]	Loss:0.001172	29.8min
[78/500]	Loss:0.001193	30.1min
[79/500]	Loss:0.001175	30.4min
[80/500]	Loss:0.001270	30.7min
[81/500]	Loss:0.001123	31.0min
[82/500]	Loss:0.001216	31.3min

[83/500]	Loss:0.001251	31.6min
[84/500]	Loss:0.001124	31.9min
[85/500]	Loss:0.001229	32.2min
[86/500]	Loss:0.001129	32.5min
[87/500]	Loss:0.001188	32.8min
[88/500]	Loss:0.001197	33.1min
[89/500]	Loss:0.001205	33.4min
[90/500]	Loss:0.001131	33.7min
[91/500]	Loss:0.001114	34.0min
[92/500]	Loss:0.001166	34.3min
[93/500]	Loss:0.001150	34.6min
[94/500]	Loss:0.001236	34.9min
[95/500]	Loss:0.001197	35.2min
[96/500]	Loss:0.001100	35.5min
[97/500]	Loss:0.001098	35.8min
[98/500]	Loss:0.001217	36.1min
[99/500]	Loss:0.001100	36.4min
[100/500]	Loss:0.001183	36.7min
[101/500]	Loss:0.001129	37.0min
[102/500]	Loss:0.001089	37.3min
[103/500]	Loss:0.001214	37.6min
[104/500]	Loss:0.001149	37.9min
[105/500]	Loss:0.001182	38.2min
[106/500]	Loss:0.001120	38.5min
[107/500]	Loss:0.001126	38.8min
[108/500]	Loss:0.001180	39.1min
[109/500]	Loss:0.001116	39.4min
[110/500]	Loss:0.001139	39.7min
[111/500]	Loss:0.001069	40.0min
[112/500]	Loss:0.001151	40.3min
[113/500]	Loss:0.001188	40.6min
[114/500]	Loss:0.001128	40.9min
[115/500]	Loss:0.001153	41.2min
[116/500]	Loss:0.001074	41.5min
[117/500]	Loss:0.001086	41.8min
[118/500]	Loss:0.001046	42.1min
[119/500]	Loss:0.001165	42.4min
[120/500]	Loss:0.001188	42.7min
[121/500]	Loss:0.001098	43.0min
[122/500]	Loss:0.001175	43.3min
[123/500]	Loss:0.001126	43.6min
[124/500]	Loss:0.001150	43.9min
[125/500]	Loss:0.001073	44.2min
[126/500]	Loss:0.001166	44.5min
[127/500]	Loss:0.001132	44.8min
[128/500]	Loss:0.001147	45.1min
[129/500]	Loss:0.001093	45.4min
[130/500]	Loss:0.001030	45.7min
[131/500]	Loss:0.001138	46.0min

[132/500]	Loss:0.001011	46.3min
[133/500]	Loss:0.001038	46.6min
[134/500]	Loss:0.001126	46.9min
[135/500]	Loss:0.001159	47.2min
[136/500]	Loss:0.001094	47.5min
[137/500]	Loss:0.001082	47.8min
[138/500]	Loss:0.001086	48.1min
[139/500]	Loss:0.001042	48.4min
[140/500]	Loss:0.001088	48.7min
[141/500]	Loss:0.001144	49.0min
[142/500]	Loss:0.001063	49.3min
[143/500]	Loss:0.001053	49.6min
[144/500]	Loss:0.001043	49.9min
[145/500]	Loss:0.001072	50.2min
[146/500]	Loss:0.001081	50.5min
[147/500]	Loss:0.001002	50.8min
[148/500]	Loss:0.001003	51.1min
[149/500]	Loss:0.001074	51.4min
[150/500]	Loss:0.000989	51.7min
[151/500]	Loss:0.000978	52.0min
[152/500]	Loss:0.001036	52.3min
[153/500]	Loss:0.001061	52.6min
[154/500]	Loss:0.000975	52.9min
[155/500]	Loss:0.001020	53.2min
[156/500]	Loss:0.001033	53.5min
[157/500]	Loss:0.001052	53.8min
[158/500]	Loss:0.001059	54.1min
[159/500]	Loss:0.001069	54.4min
[160/500]	Loss:0.000995	54.7min
[161/500]	Loss:0.001108	55.0min
[162/500]	Loss:0.001001	55.3min
[163/500]	Loss:0.001076	55.6min
[164/500]	Loss:0.001082	55.9min
[165/500]	Loss:0.000986	56.2min
[166/500]	Loss:0.000942	56.5min
[167/500]	Loss:0.000975	56.8min
[168/500]	Loss:0.001021	57.1min
[169/500]	Loss:0.000967	57.4min
[170/500]	Loss:0.001053	57.7min
[171/500]	Loss:0.000947	58.0min
[172/500]	Loss:0.000951	58.3min
[173/500]	Loss:0.000946	58.6min
[174/500]	Loss:0.001001	58.9min
[175/500]	Loss:0.001060	59.2min
[176/500]	Loss:0.000998	59.5min
[177/500]	Loss:0.000935	59.8min
[178/500]	Loss:0.001033	60.1min
[179/500]	Loss:0.000994	60.4min
[180/500]	Loss:0.001024	60.7min

[181/500]	Loss:0.000958	61.0min
[182/500]	Loss:0.001014	61.3min
[183/500]	Loss:0.001027	61.6min
[184/500]	Loss:0.000977	61.9min
[185/500]	Loss:0.000929	62.2min
[186/500]	Loss:0.000942	62.5min
[187/500]	Loss:0.001038	62.8min
[188/500]	Loss:0.000921	63.1min
[189/500]	Loss:0.000929	63.4min
[190/500]	Loss:0.001043	63.7min
[191/500]	Loss:0.000993	64.0min
[192/500]	Loss:0.000971	64.3min
[193/500]	Loss:0.001027	64.6min
[194/500]	Loss:0.000987	64.9min
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[197/500]	Loss:0.000952	65.8min
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[199/500]	Loss:0.001035	66.4min
[200/500]	Loss:0.000951	66.7min
[201/500]	Loss:0.000933	67.0min
[202/500]	Loss:0.000870	67.3min
[203/500]	Loss:0.000920	67.6min
[204/500]	Loss:0.001020	67.9min
[205/500]	Loss:0.000962	68.2min
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[207/500]	Loss:0.000896	68.8min
[208/500]	Loss:0.001004	69.1min
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[210/500]	Loss:0.000989	69.7min
[211/500]	Loss:0.000860	70.0min
[212/500]	Loss:0.000871	70.3min
[213/500]	Loss:0.000935	70.6min
[214/500]	Loss:0.000887	70.9min
[215/500]	Loss:0.000925	71.2min
[216/500]	Loss:0.000969	71.5min
[217/500]	Loss:0.000993	71.8min
[218/500]	Loss:0.000995	72.1min
[219/500]	Loss:0.000915	72.4min
[220/500]	Loss:0.000940	72.7min
[221/500]	Loss:0.000891	73.0min
[222/500]	Loss:0.000852	73.3min
[223/500]	Loss:0.000828	73.6min
[224/500]	Loss:0.000914	73.9min
[225/500]	Loss:0.000904	74.2min
[226/500]	Loss:0.000876	74.5min
[227/500]	Loss:0.000971	74.8min
[228/500]	Loss:0.000915	75.1min
[229/500]	Loss:0.000817	75.4min

[230/500]	Loss:0.000905	75.7min
[231/500]	Loss:0.000854	76.0min
[232/500]	Loss:0.000942	76.3min
[233/500]	Loss:0.000901	76.6min
[234/500]	Loss:0.000827	76.9min
[235/500]	Loss:0.000953	77.2min
[236/500]	Loss:0.000833	77.5min
[237/500]	Loss:0.000811	77.8min
[238/500]	Loss:0.000852	78.1min
[239/500]	Loss:0.000811	78.4min
[240/500]	Loss:0.000949	78.7min
[241/500]	Loss:0.000864	79.0min
[242/500]	Loss:0.000914	79.3min
[243/500]	Loss:0.000849	79.6min
[244/500]	Loss:0.000925	79.9min
[245/500]	Loss:0.000919	80.2min
[246/500]	Loss:0.000811	80.5min
[247/500]	Loss:0.000837	80.8min
[248/500]	Loss:0.000824	81.1min
[249/500]	Loss:0.000870	81.4min
[250/500]	Loss:0.000904	81.7min
[251/500]	Loss:0.000932	82.0min
[252/500]	Loss:0.000907	82.3min
[253/500]	Loss:0.000822	82.6min
[254/500]	Loss:0.000768	82.9min
[255/500]	Loss:0.000840	83.2min
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[257/500]	Loss:0.000911	83.8min
[258/500]	Loss:0.000819	84.1min
[259/500]	Loss:0.000760	84.4min
[260/500]	Loss:0.000864	84.7min
[261/500]	Loss:0.000849	85.0min
[262/500]	Loss:0.000789	85.3min
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[276/500]	Loss:0.000747	89.5min
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[279/500]	Loss:0.000846	90.4min
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[294/500]	Loss:0.000834	94.9min
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[303/500]	Loss:0.000817	97.6min
[304/500]	Loss:0.000675	97.9min
[305/500]	Loss:0.000711	98.2min
[306/500]	Loss:0.000814	98.5min
[307/500]	Loss:0.000770	98.8min
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[309/500]	Loss:0.000770	99.4min
[310/500]	Loss:0.000734	99.7min
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[316/500]	Loss:0.000750	101.5min
[317/500]	Loss:0.000658	101.8min
[318/500]	Loss:0.000774	102.1min
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[320/500]	Loss:0.000670	102.7min
[321/500]	Loss:0.000705	103.0min
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[323/500]	Loss:0.000645	103.6min
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[325/500]	Loss:0.000742	104.2min
[326/500]	Loss:0.000671	104.5min
[327/500]	Loss:0.000719	104.8min

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[329/500]	Loss:0.000704	105.4min
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[332/500]	Loss:0.000620	106.3min
[333/500]	Loss:0.000665	106.6min
[334/500]	Loss:0.000663	106.9min
[335/500]	Loss:0.000727	107.2min
[336/500]	Loss:0.000607	107.5min
[337/500]	Loss:0.000624	107.8min
[338/500]	Loss:0.000700	108.1min
[339/500]	Loss:0.000640	108.4min
[340/500]	Loss:0.000650	108.7min
[341/500]	Loss:0.000697	109.0min
[342/500]	Loss:0.000701	109.3min
[343/500]	Loss:0.000706	109.6min
[344/500]	Loss:0.000629	109.9min
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[346/500]	Loss:0.000649	110.5min
[347/500]	Loss:0.000622	110.8min
[348/500]	Loss:0.000707	111.1min
[349/500]	Loss:0.000613	111.4min
[350/500]	Loss:0.000730	111.7min
[351/500]	Loss:0.000666	112.0min
[352/500]	Loss:0.000622	112.3min
[353/500]	Loss:0.000644	112.6min
[354/500]	Loss:0.000713	112.9min
[355/500]	Loss:0.000726	113.2min
[356/500]	Loss:0.000622	113.5min
[357/500]	Loss:0.000578	113.8min
[358/500]	Loss:0.000663	114.1min
[359/500]	Loss:0.000676	114.4min
[360/500]	Loss:0.000580	114.7min
[361/500]	Loss:0.000663	115.0min
[362/500]	Loss:0.000554	115.3min
[363/500]	Loss:0.000595	115.6min
[364/500]	Loss:0.000704	115.9min
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[368/500]	Loss:0.000559	117.1min
[369/500]	Loss:0.000658	117.4min
[370/500]	Loss:0.000560	117.7min
[371/500]	Loss:0.000593	118.0min
[372/500]	Loss:0.000535	118.3min
[373/500]	Loss:0.000588	118.6min
[374/500]	Loss:0.000611	118.9min
[375/500]	Loss:0.000625	119.2min
[376/500]	Loss:0.000543	119.5min

[377/500]	Loss:0.000569		119.8min
[378/500]	Loss:0.000645		120.1min
[379/500]	Loss:0.000580		120.4min
[380/500]	Loss:0.000658		120.7min
[381/500]	Loss:0.000556		121.0min
[382/500]	Loss:0.000546		121.3min
[383/500]	Loss:0.000615		121.6min
[384/500]	Loss:0.000582		121.9min
[385/500]	Loss:0.000601		122.2min
[386/500]	Loss:0.000635		122.5min
[387/500]	Loss:0.000636		122.8min
[388/500]	Loss:0.000609		123.1min
[389/500]	Loss:0.000612		123.4min
[390/500]	Loss:0.000635		123.7min
[391/500]	Loss:0.000618		124.0min
[392/500]	Loss:0.000521		124.3min
[393/500]	Loss:0.000522		124.6min
[394/500]	Loss:0.000584		124.9min
[395/500]	Loss:0.000582		125.2min
[396/500]	Loss:0.000542		125.5min
[397/500]	Loss:0.000509		125.8min
[398/500]	Loss:0.000533		126.1min
[399/500]	Loss:0.000575		126.4min
[400/500]	Loss:0.000504		126.7min
[401/500]	Loss:0.000604		127.0min
[402/500]	Loss:0.000573		127.3min
[403/500]	Loss:0.000479		127.6min
[404/500]	Loss:0.000519		127.9min
[405/500]	Loss:0.000515		128.2min
[406/500]	Loss:0.000551		128.5min
[407/500]	Loss:0.000578		128.8min
[408/500]	Loss:0.000485		129.1min
[409/500]	Loss:0.000476		129.4min
[410/500]	Loss:0.000547		129.7min
[411/500]	Loss:0.000556		130.0min
[412/500]	Loss:0.000521		130.3min
[413/500]	Loss:0.000515		130.6min
[414/500]	Loss:0.000521		130.9min
[415/500]	Loss:0.000580		131.2min
[416/500]	Loss:0.000535		131.5min
[417/500]	Loss:0.000527		131.8min
[418/500]	Loss:0.000454		132.1min
[419/500]	Loss:0.000463		132.4min
[420/500]	Loss:0.000593		132.7min
[421/500]	Loss:0.000480		133.0min
[422/500]	Loss:0.000478		133.3min
[423/500]	Loss:0.000557		133.6min
[424/500]	Loss:0.000479		133.9min
[425/500]	Loss:0.000586		134.2min

```
[426/500] Loss:0.000452 | 134.5min
[427/500] Loss:0.000494 | 134.8min
[428/500] Loss:0.000568 | 135.1min
[429/500] Loss:0.000518 | 135.4min
[430/500] Loss:0.000547 | 135.7min
[431/500] Loss:0.000521 | 136.0min
[432/500] Loss:0.000438 | 136.3min
[433/500] Loss:0.000455 | 136.6min
[434/500] Loss:0.000534 | 136.9min
[435/500] Loss:0.000508 | 137.2min
[436/500] Loss:0.000447 | 137.5min
[437/500] Loss:0.000547 | 137.8min
[438/500] Loss:0.000505 | 138.1min
[439/500] Loss:0.000496 | 138.4min
[440/500] Loss:0.000470 | 138.7min
[441/500] Loss:0.000504 | 139.0min
[442/500] Loss:0.000481 | 139.3min
[443/500] Loss:0.000526 | 139.6min
[444/500] Loss:0.000435 | 139.9min
[445/500] Loss:0.000513 | 140.2min
[446/500] Loss:0.000509 | 140.5min
[447/500] Loss:0.000480 | 140.8min
[448/500] Loss:0.000469 | 141.1min
[449/500] Loss:0.000435 | 141.4min
[450/500] Loss:0.000537 | 141.7min
[451/500] Loss:0.000435 | 142.0min
[452/500] Loss:0.000494 | 142.3min
[453/500] Loss:0.000528 | 142.6min
[454/500] Loss:0.000435 | 142.9min
[455/500] Loss:0.000465 | 143.2min
[456/500] Loss:0.000448 | 143.5min
[457/500] Loss:0.000445 | 143.8min
[458/500] Loss:0.000470 | 144.1min
[459/500] Loss:0.000435 | 144.4min
[460/500] Loss:0.000435 | 144.7min
[461/500] Loss:0.000511 | 145.0min
[462/500] Loss:0.000435 | 145.3min
```

Early stopping at epoch 462

Losses saved to:

/content/drive/MyDrive/DLENDSEM/final_v2/results/losses.json

Cell 9: Evaluate Part A & Part B

```
PLAN1 = '/content/drive/MyDrive/DLENDSEM/plan1_csrnet'
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
SHA = '/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech'
```

```

tf_eval = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406], [0.229,0.224,0.225])
])

def evaluate(model, test_imgs, h5_dir, label):
    model.eval()
    mae_s, mse_s, results = 0, 0, []
    with torch.no_grad():
        for i, ip in enumerate(test_imgs):
            fname = os.path.basename(ip).replace('.jpg', '')
            img = Image.open(ip).convert('RGB')
            inp = tf_eval(img).unsqueeze(0).cuda()
            h5p = f'{h5_dir}/{fname}.h5'
            with h5py.File(h5p, 'r') as f:
                gt = float(np.array(f['density']).sum())
            pred = model(inp).sum().item()
            err = abs(pred - gt)
            mae_s += err
            mse_s += err**2
            results.append({'file':fname, 'gt':round(gt,2),
                'pred':round(pred,2), 'err':round(err,2)})
            if (i+1) % 60 == 0:
                print(f' [{label}] {i+1}/{len(test_imgs)}')
        n = len(test_imgs)
        mae, mse = mae_s/n, (mse_s/n)**0.5
        print(f' [{label}]: MAE={mae:.2f} MSE={mse:.2f}')
    return {'mae':round(mae,2), 'mse':round(mse,2), 'n':n,
        'per_image':results}

test_A = sorted(glob.glob(f'{SHA}/part_A/test_data/images/*.jpg'))
test_B = sorted(glob.glob(f'{SHA}/part_B/test_data/images/*.jpg'))

print('Evaluating...')
res_A = evaluate(model, test_A, f'{FINAL}/density_maps_test_A', 'Part A')
res_B = evaluate(model, test_B, f'{FINAL}/density_maps_test_B', 'Part B')

with open(f'{FINAL}/results/eval.json', 'w') as f:
    json.dump({'partA':res_A, 'partB':res_B}, f, indent=2)
print('Saved!')

Evaluating...
[Part A] 60/182
[Part A] 120/182
[Part A] 180/182
Part A: MAE=63.31 MSE=109.62
[Part B] 60/316
[Part B] 120/316

```

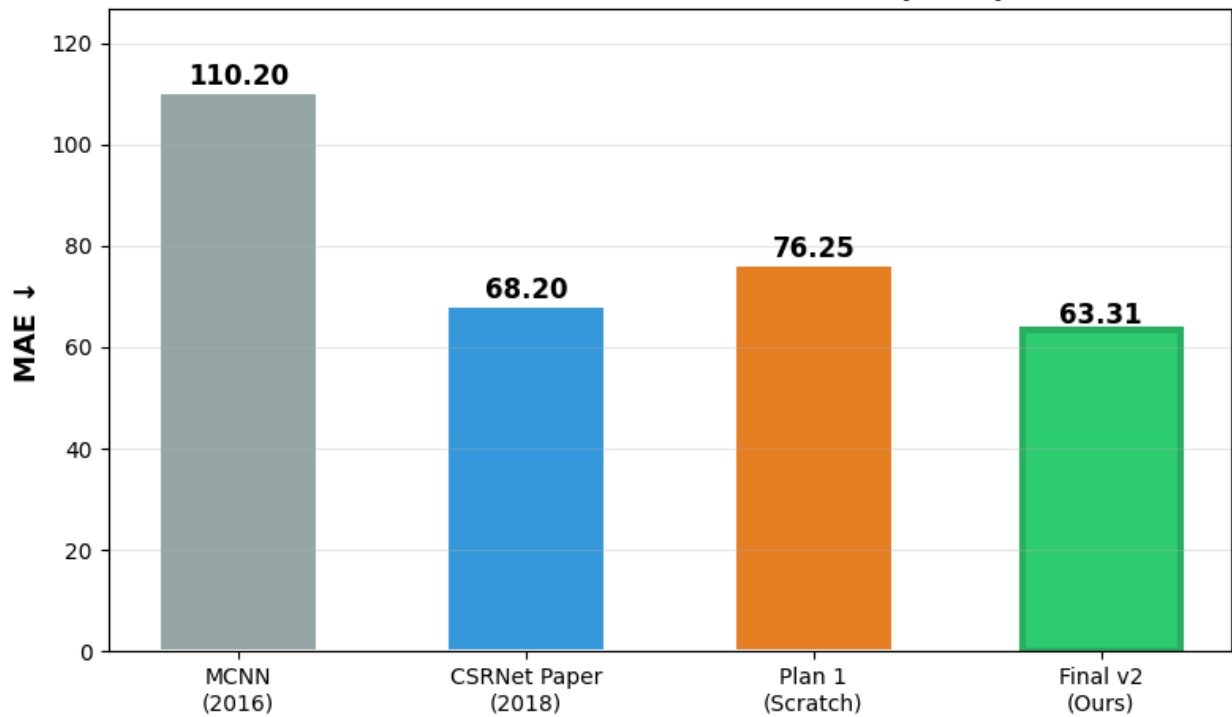
```
[Part B] 180/316
[Part B] 240/316
[Part B] 300/316
Part B: MAE=8.37 MSE=13.38
Saved!
```

```
import matplotlib.pyplot as plt
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'

models = ['MCNN\n(2016)', 'CSRNet Paper\n(2018)', 'Plan 1\n(Scratch)',
'Final v2\n(Ours)']
vals = [110.2, 68.2, 76.25, 63.31]
colors = ['#95a5a6', '#3498db', '#e67e22', '#2ecc71']

fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(models, vals, color=colors, edgecolor='white',
linewidth=1.5, width=0.55)
bars[-1].set_edgecolor('#27ae60'); bars[-1].set_linewidth(3)
for b, v in zip(bars, vals):
    ax.text(b.get_x()+b.get_width()/2, v+1.5, f'{v:.2f}', ha='center',
    fontsize=12, fontweight='bold')
ax.set_ylabel('MAE ↓', fontsize=13, fontweight='bold')
ax.set_title('Part A – Mean Absolute Error (MAE)', fontsize=15,
fontweight='bold')
ax.set_ylim(0, max(vals)*1.15); ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig(f'{FINAL}/visualizations/mae_partA.png', dpi=200,
bbox_inches='tight')
plt.show()
```


Part A — Mean Absolute Error (MAE)

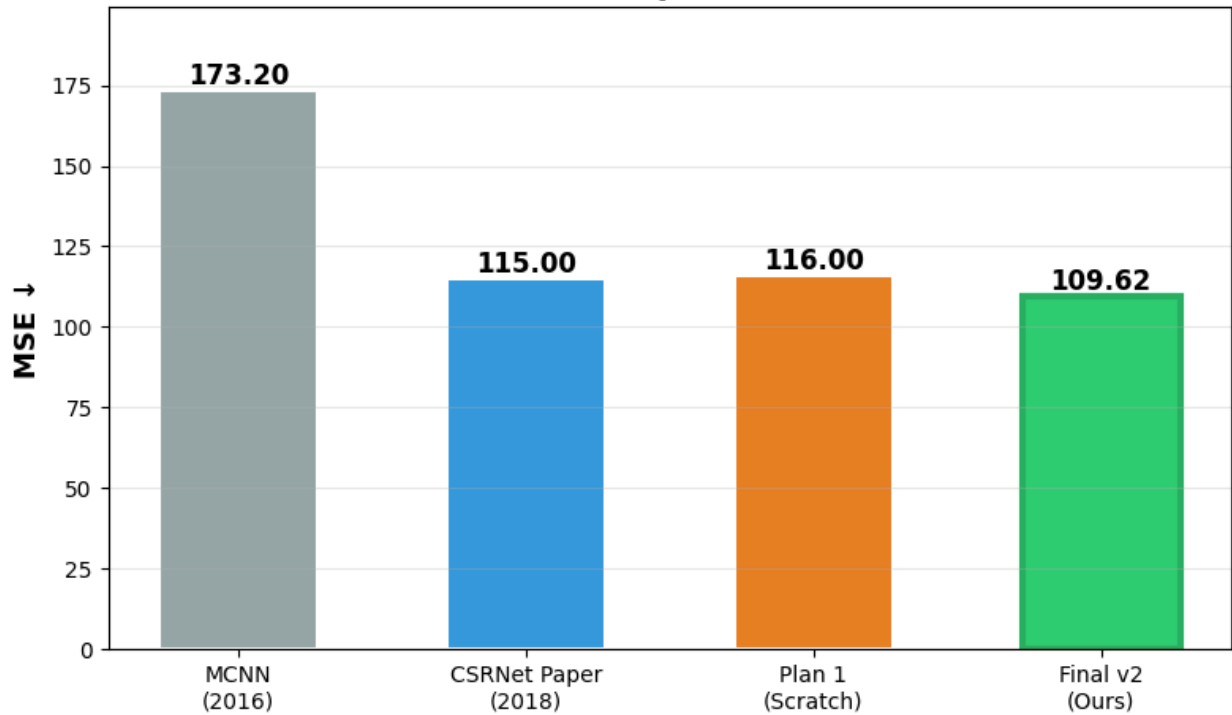


```
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'

models = ['MCNN\n(2016)', 'CSRNet Paper\n(2018)', 'Plan 1\n(Scratch)',
'Final v2\n(Ours)']
vals = [173.2, 115.0, 116.0, 109.62]
colors = ['#95a5a6', '#3498db', '#e67e22', '#2ecc71']

fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(models, vals, color=colors, edgecolor='white',
linewidth=1.5, width=0.55)
bars[-1].set_edgecolor('#27ae60'); bars[-1].set_linewidth(3)
for b, v in zip(bars, vals):
    ax.text(b.get_x()+b.get_width()/2, v+2, f'{v:.2f}', ha='center',
    fontsize=12, fontweight='bold')
ax.set_ylabel('MSE ↓', fontsize=13, fontweight='bold')
ax.set_title('Part A — Mean Squared Error (MSE)', fontsize=15,
fontweight='bold')
ax.set_ylim(0, max(vals)*1.15); ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig(f'{FINAL}/visualizations/mse_partA.png', dpi=200,
bbox_inches='tight')
plt.show()
```

Part A — Mean Squared Error (MSE)



```
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
```

```
models = ['MCNN\n(2016)', 'CSRNet Paper\n(2018)', 'Final v2\n(Ours)']
```

```
vals = [26.4, 10.6, 8.37]
```

```
colors = ['#95a5a6', '#3498db', '#2ecc71']
```

```
fig, ax = plt.subplots(figsize=(7, 5))
```

```
bars = ax.bar(models, vals, color=colors, edgecolor='white',  
linewidth=1.5, width=0.5)
```

```
bars[-1].set_edgecolor('#27ae60'); bars[-1].set_linewidth(3)
```

```
for b, v in zip(bars, vals):
```

```
    ax.text(b.get_x()+b.get_width()/2, v+0.5, f'{v:.2f}', ha='center',  
    fontsize=12, fontweight='bold')
```

```
ax.set_ylabel('MAE ↓', fontsize=13, fontweight='bold')
```

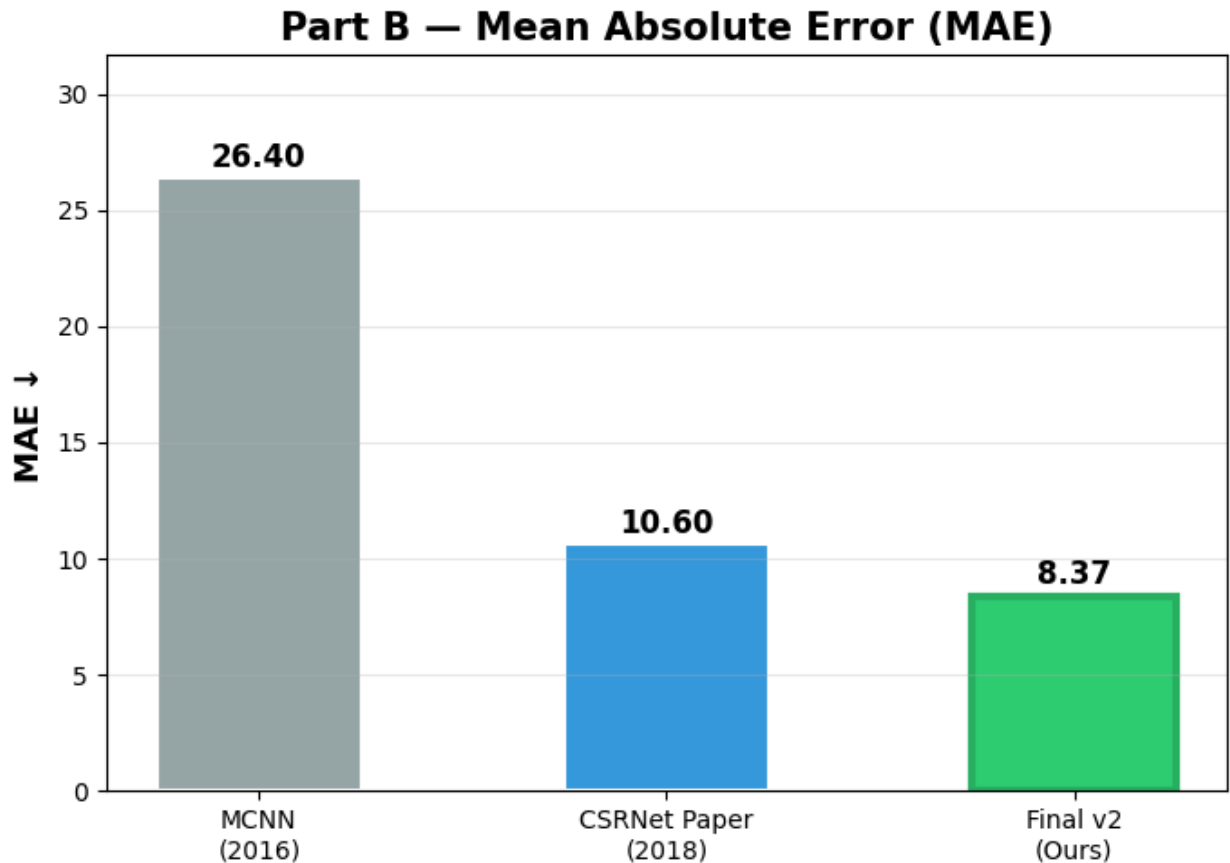
```
ax.set_title('Part B — Mean Absolute Error (MAE)', fontsize=15,  
fontweight='bold')
```

```
ax.set_ylim(0, max(vals)*1.2); ax.grid(axis='y', alpha=0.3)
```

```
plt.tight_layout()
```

```
plt.savefig(f'{FINAL}/visualizations/mae_partB.png', dpi=200,  
bbox_inches='tight')
```

```
plt.show()
```

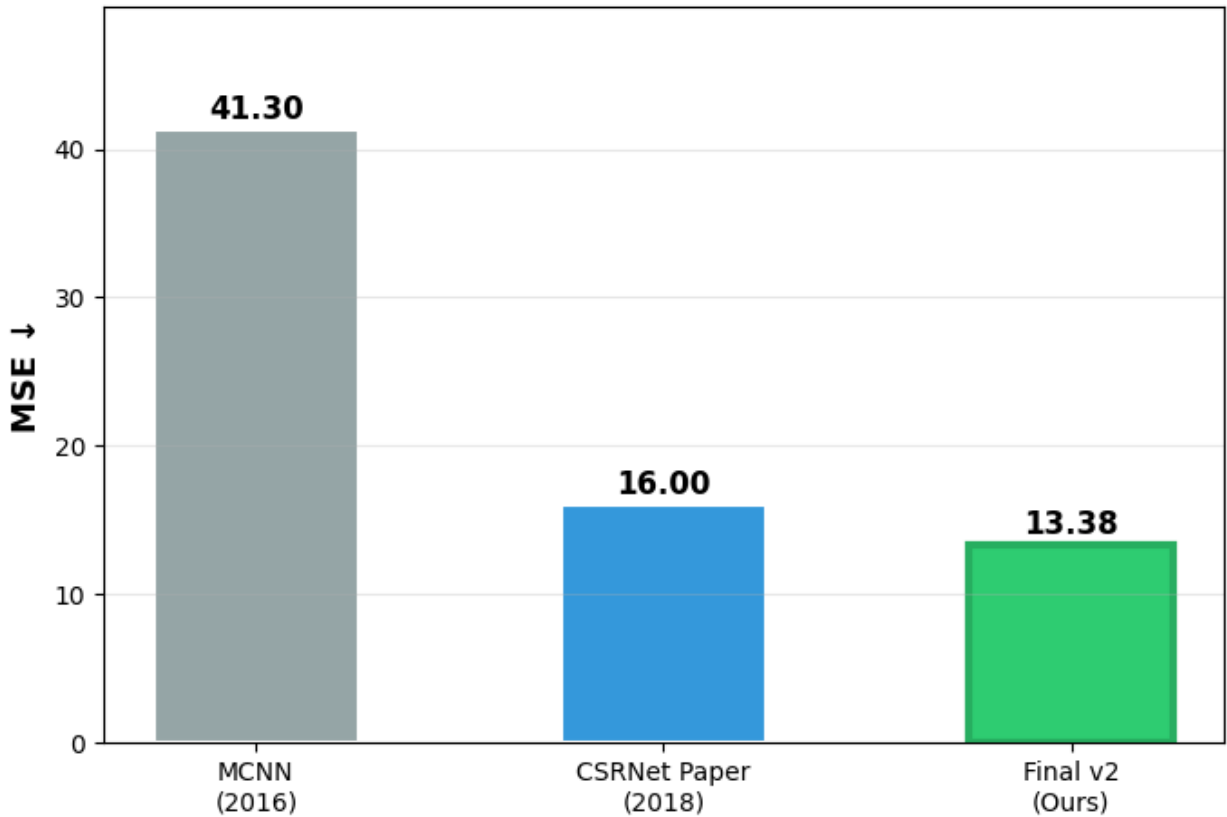


```
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'

models = ['MCNN\n(2016)', 'CSRNet Paper\n(2018)', 'Final v2\n(Ours)']
vals = [41.3, 16.0, 13.38]
colors = ['#95a5a6', '#3498db', '#2ecc71']

fig, ax = plt.subplots(figsize=(7, 5))
bars = ax.bar(models, vals, color=colors, edgecolor='white',
linewidth=1.5, width=0.5)
bars[-1].set_edgecolor('#27ae60'); bars[-1].set_linewidth(3)
for b, v in zip(bars, vals):
    ax.text(b.get_x()+b.get_width()/2, v+0.7, f'{v:.2f}', ha='center',
    fontsize=12, fontweight='bold')
ax.set_ylabel('MSE ↓', fontsize=13, fontweight='bold')
ax.set_title('Part B — Mean Squared Error (MSE)', fontsize=15,
fontweight='bold')
ax.set_ylim(0, max(vals)*1.2); ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig(f'{FINAL}/visualizations/mse_partB.png', dpi=200,
bbox_inches='tight')
plt.show()
```

Part B — Mean Squared Error (MSE)



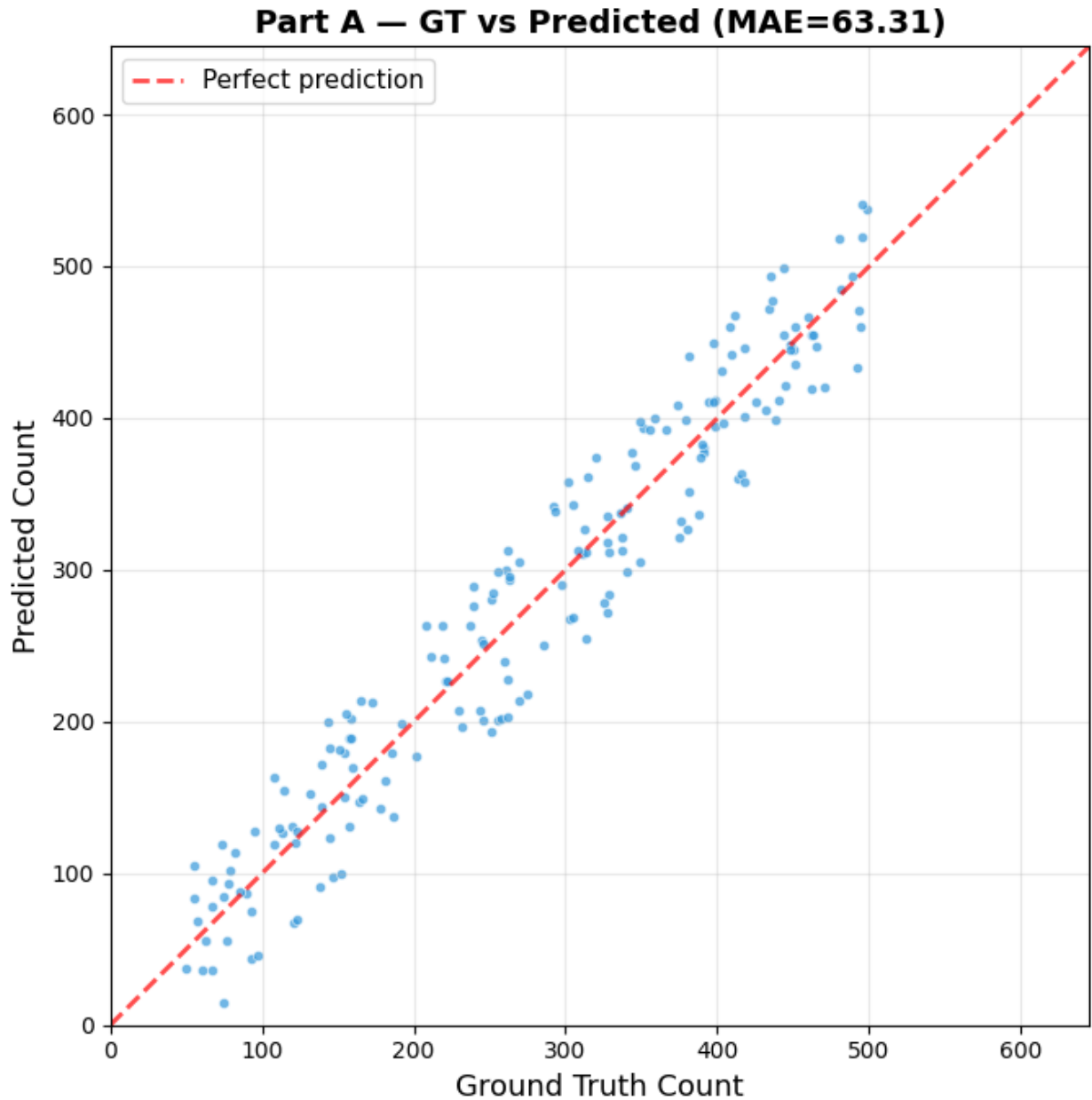
```
import json, numpy as np, matplotlib.pyplot as plt
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'

with open(f'{FINAL}/results/eval.json') as f:
    results = json.load(f)

gts = [r['gt'] for r in results['partA']['per_image']]
preds = [r['pred'] for r in results['partA']['per_image']]
mx = max(max(gts), max(preds)) * 1.1

fig, ax = plt.subplots(figsize=(7, 7))
ax.scatter(gts, preds, c='#3498db', alpha=0.5, s=40,
           edgecolors='white', linewidth=0.5)
ax.plot([0, mx], [0, mx], 'r--', linewidth=2, alpha=0.7,
        label='Perfect prediction')
ax.set_xlabel('Ground Truth Count', fontsize=13)
ax.set_ylabel('Predicted Count', fontsize=13)
ax.set_title(f'Part A — GT vs Predicted (MAE={results["partA"]\n["mae"]})', fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
ax.set_xlim(0, mx); ax.set_ylim(0, mx)
ax.set_aspect('equal')
ax.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.savefig(f'{FINAL}/visualizations/scatter_partA.png', dpi=200,
bbox_inches='tight')
plt.show()
```



```
FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
with open(f'{FINAL}/results/eval.json') as f:
    results = json.load(f)
gts = [r['gt'] for r in results['partB']['per_image']]
```

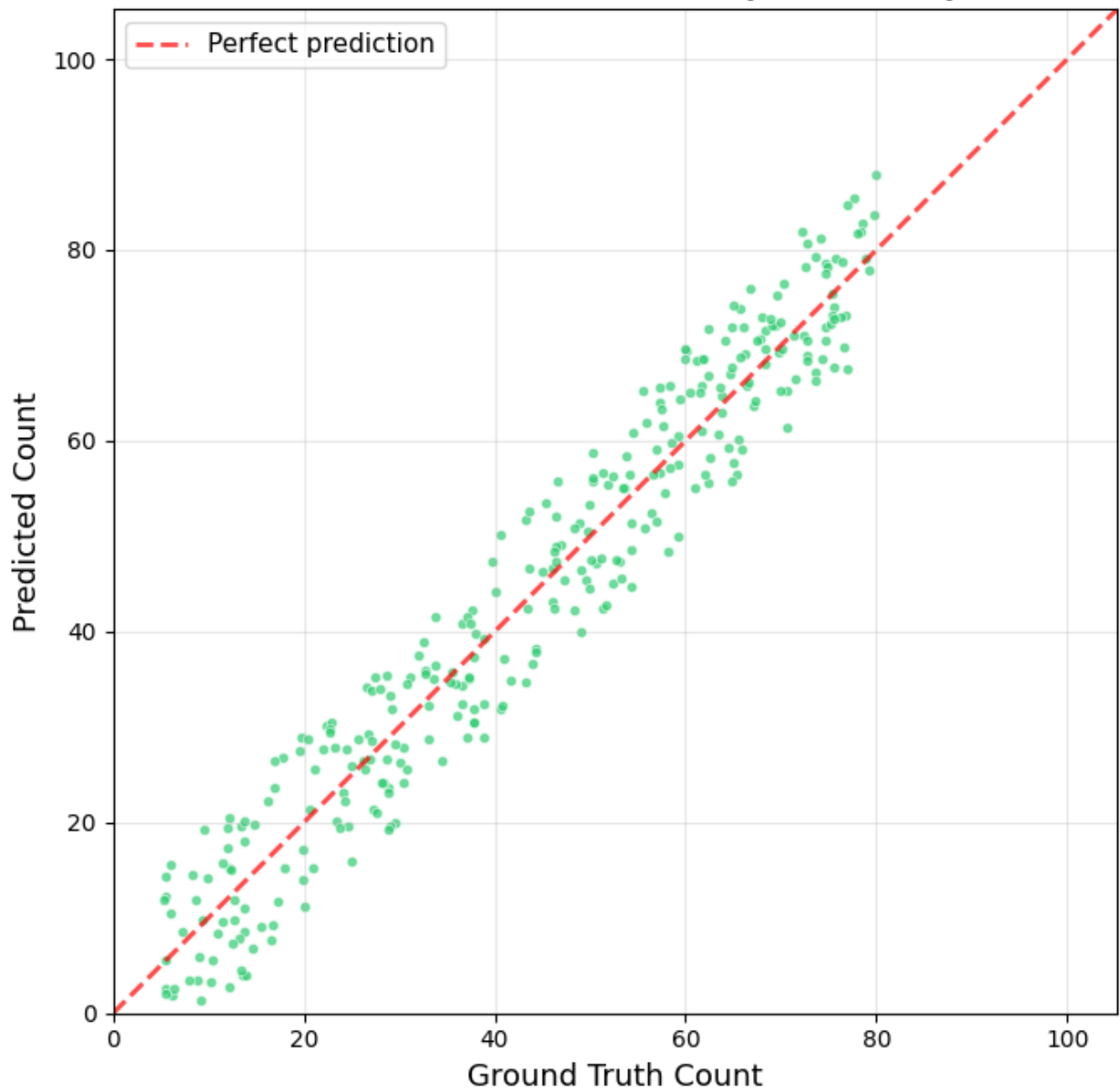
```

preds = [r['pred'] for r in results['partB']['per_image']]
mx = max(max(gts), max(preds)) * 1.1

fig, ax = plt.subplots(figsize=(7, 7))
ax.scatter(gts, preds, c='#2ecc71', alpha=0.5, s=40,
edgecolors='white', linewidth=0.5)
ax.plot([0, mx], [0, mx], 'r--', linewidth=2, alpha=0.7,
label='Perfect prediction')
ax.set_xlabel('Ground Truth Count', fontsize=13)
ax.set_ylabel('Predicted Count', fontsize=13)
ax.set_title(f'Part B – GT vs Predicted (MAE={results["partB"]
["mae"]})', fontsize=14, fontweight='bold')
ax.legend(fontsize=11)
ax.set_xlim(0, mx); ax.set_ylim(0, mx)
ax.set_aspect('equal')
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig(f'{FINAL}/visualizations/scatter_partB.png', dpi=200,
bbox_inches='tight')
plt.show()

```

Part B — GT vs Predicted (MAE=8.37)



```
import torch, torch.nn as nn, glob, os, h5py, numpy as np
from torchvision import transforms, models
from PIL import Image
import matplotlib.pyplot as plt

FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
C_DIR =
'/content/drive/MyDrive/DLENDSEM/dataset/ShanghaiTech/part_C_india'

class CSRNet(nn.Module):
    def __init__(self):
        super().__init__()
```

```

vgg = models.vgg16(weights=models.VGG16_Weights.IMAGENET1K_V1)
self.frontend = nn.Sequential(*list(vgg.features.children())
[:23])
self.backend = nn.Sequential(
    nn.Conv2d(512,512,3,padding=2,dilation=2), nn.ReLU(True),
    nn.Conv2d(512,512,3,padding=2,dilation=2), nn.ReLU(True),
    nn.Conv2d(512,512,3,padding=2,dilation=2), nn.ReLU(True),
    nn.Conv2d(512,256,3,padding=2,dilation=2), nn.ReLU(True),
    nn.Conv2d(256,128,3,padding=2,dilation=2), nn.ReLU(True),
    nn.Conv2d(128,64, 3,padding=2,dilation=2), nn.ReLU(True))
self.output_layer = nn.Conv2d(64, 1, 1)

def forward(self, x):
    return self.output_layer(self.backend(self.frontend(x)))

model = CSRNet().cuda()
ckpt = torch.load(PRETRAINED, map_location='cpu', weights_only=False)
if isinstance(ckpt, dict) and 'state_dict' in ckpt:
    state = {k.replace('module.', ''):v for k,v in
ckpt['state_dict'].items()}
else:
    state = {k.replace('module.', ''):v for k,v in ckpt.items()}
model.load_state_dict(state, strict=False)
model.eval()

tf_eval = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406], [0.229,0.224,0.225])
])

test_C_imgs = sorted(glob.glob(f'{C_DIR}/test_data/images/*.jpg'))
d_map_dir_C = f'{FINAL}/density_maps_test_C'

print(f'Making predictions on {len(test_C_imgs)} Part C test
images...')

fig, axes = plt.subplots(2, 5, figsize=(25, 6))
fig.suptitle('Part C Test Images - Ground Truth vs. Predicted Count',
fontsize=16, fontweight='bold')

for i, ax in enumerate(axes.flat):
    if i >= len(test_C_imgs):
        ax.axis('off')
        continue

    img_path = test_C_imgs[i]
    fname = os.path.basename(img_path).replace('.jpg', '')

    img = Image.open(img_path).convert('RGB')

```



```

gt_path = f'{d_map_dir_C}/{fname}.h5'
gt_count = 0.0
if os.path.exists(gt_path):
    with h5py.File(gt_path, 'r') as f:
        gt_count = np.array(f['density']).sum()
else:
    print(f"Warning: Ground truth density map not found for {fname}. GT count will be 0.")

input_tensor = tf_eval(img).unsqueeze(0).cuda()
with torch.no_grad():
    output = model(input_tensor)
    pred_count = output.sum().item()

ax.imshow(img)
ax.set_title(f'GT: {gt_count:.0f} | Pred: {pred_count:.0f}',
            fontsize=12, fontweight='bold')
ax.axis('off')

plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.savefig(f'{FINAL}/visualizations/partC_predictions_sample.png',
            dpi=200, bbox_inches='tight')
plt.show()

```

Making predictions on 10 Part C test images...



Cell 10: Results Comparison

```

print('='*62)
print(f"{'Model':<38} {'MAE':>8} {'MSE':>8} Split")
print('='*62)
rows=[
    ('MCNN (2016)',110.2,173.2,'A'),
    ('CSRNet paper (2018)',68.2,115.0,'A'),
    ('CSRNet paper (2018)',10.6,16.0,'B'),
    ('Plan1 - our baseline',76.25,116.0,'A'),
    ('FINAL (baseline pretrained+ft)',res_A['mae'],res_A['mse'],'A'),
    ('FINAL (baseline pretrained+ft)',res_B['mae'],res_B['mse'],'B')]
for name,mae,mse,sp in rows:

```

```

tag=' <-- Final' if 'FINAL' in name else ''
print(f' {name:<36} {mae:>8.2f} {mse:>8.2f} [Part {sp}]{tag}')
print('='*62)

```

```

=====
Model                                     MAE      MSE  Split
=====
MCNN (2016)                             110.20   173.20 [Part A]
CSRNet paper (2018)                     68.20    115.00 [Part A]
CSRNet paper (2018)                     10.60     16.00 [Part B]
Plan1 - our baseline                     76.25    116.00 [Part A]
FINAL (baseline pretrained+ft)          63.31    109.62 [Part A] <--
Final
FINAL (baseline pretrained+ft)           8.37     13.38 [Part B] <--
Final
=====

```

```

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

FINAL = '/content/drive/MyDrive/DLENDSEM/final_v2'
fig, ax = plt.subplots(figsize=(18, 6))
ax.set_xlim(0, 18); ax.set_ylim(0, 6); ax.axis('off')
ax.set_title('CSRNet Architecture – VGG-16 Frontend + Dilated
Backend', fontsize=16, fontweight='bold', pad=20)

def draw_block(ax, x, y, w, h, label, sublabel, color,
textcolor='white'):
    rect = mpatches.FancyBboxPatch((x, y), w, h,
boxstyle="round,pad=0.1",
                                facecolor=color,
edgecolor='white', linewidth=2)
    ax.add_patch(rect)
    ax.text(x+w/2, y+h/2+0.15, label, ha='center', va='center',
fontsize=9, fontweight='bold', color=textcolor)
    ax.text(x+w/2, y+h/2-0.25, sublabel, ha='center', va='center',
fontsize=7, color=textcolor, alpha=0.85)

def arrow(ax, x1, x2, y):
    ax.annotate('', xy=(x2, y), xytext=(x1, y),
arrowprops=dict(arrowstyle='->', color='#555', lw=2))

# Input
draw_block(ax, 0.2, 2, 1.2, 2, 'Input\nImage', 'H×W×3', '#1abc9c')

# Frontend (VGG-16 first 13 conv layers)
vgg_blocks = [
    ('Conv×2', '64', '#3498db'),
    ('Pool', '↓2×', '#2c3e50'),
    ('Conv×2', '128', '#3498db'),

```

```

        ('Pool', '↓2×', '#2c3e50'),
        ('Conv×3', '256', '#3498db'),
        ('Pool', '↓2×', '#2c3e50'),
        ('Conv×3', '512', '#3498db'),
    ]
    x = 1.8
    for label, sub, col in vgg_blocks:
        w = 0.9 if 'Pool' not in label else 0.6
        draw_block(ax, x, 2, w, 2, label, sub, col)
        arrow(ax, x-0.15, x, 3)
        x += w + 0.1

    # Frontend label
    ax.add_patch(mpatches.FancyBboxPatch((1.7, 0.5), x-1.8, 0.8,
        boxstyle="round,pad=0.1",
        facecolor='none', edgecolor='#3498db', linewidth=2,
        linestyle='--'))
    ax.text((1.7+x)/2, 0.9, 'Frontend (VGG-16, Frozen)', ha='center',
        fontsize=10, color='#3498db', fontweight='bold')

    # Backend (dilated convolutions)
    arrow(ax, x-0.15, x+0.1, 3)
    backend_blocks = [
        ('Dil Conv', '512', '#e74c3c'),
        ('Dil Conv', '512', '#e74c3c'),
        ('Dil Conv', '512', '#e74c3c'),
        ('Dil Conv', '256', '#e74c3c'),
        ('Dil Conv', '128', '#e74c3c'),
        ('Dil Conv', '64', '#e74c3c'),
    ]
    bx = x + 0.2
    for label, sub, col in backend_blocks:
        draw_block(ax, bx, 2, 0.85, 2, label, sub, col)
        arrow(ax, bx-0.15, bx, 3)
        bx += 0.95

    # Backend label
    ax.add_patch(mpatches.FancyBboxPatch((x+0.1, 0.5), bx-x-0.2, 0.8,
        boxstyle="round,pad=0.1",
        facecolor='none', edgecolor='#e74c3c', linewidth=2,
        linestyle='--'))
    ax.text((x+0.1+bx)/2, 0.9, 'Backend (Dilated Conv, d=2, Trainable)',
        ha='center', fontsize=10, color='#e74c3c', fontweight='bold')

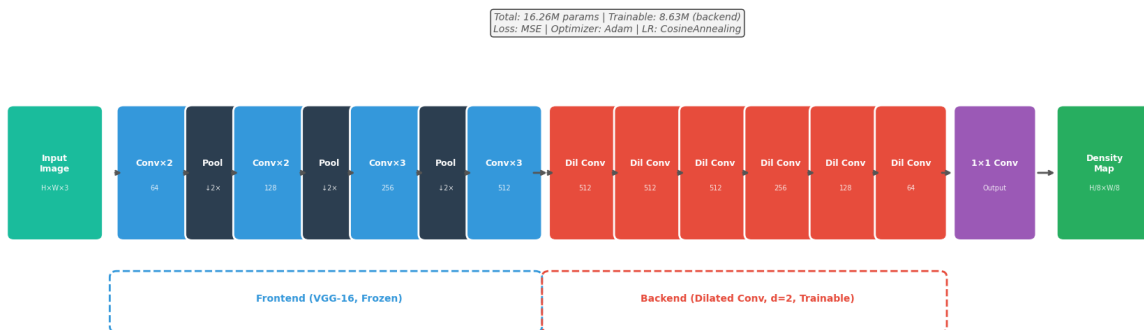
    # Output
    arrow(ax, bx-0.1, bx+0.1, 3)
    draw_block(ax, bx+0.2, 2, 1.0, 2, '1×1 Conv', 'Output', '#9b59b6')
    arrow(ax, bx+1.3, bx+1.6, 3)
    draw_block(ax, bx+1.7, 2, 1.2, 2, 'Density\nMap', 'H/8×W/8',
        '#27ae60')

```

```
# Info box
info = 'Total: 16.26M params | Trainable: 8.63M (backend)\nLoss: MSE |
Optimizer: Adam | LR: CosineAnnealing'
ax.text(9, 5.3, info, ha='center', fontsize=10, style='italic',
color='#555',
bbox=dict(boxstyle='round', facecolor='#f0f0f0', alpha=0.8))

plt.tight_layout()
plt.savefig(f'{FINAL}/visualizations/architecture.png', dpi=200,
bbox_inches='tight')
plt.show()
```

CSRNet Architecture — VGG-16 Frontend + Dilated Backend



Cell 11: Loss Curves

```
import plotly.graph_objects as go
import os
import json # Import the json library

FINAL='/content/drive/MyDrive/DLENDSEM/final_v2'

# Load losses from the JSON file
lf = f'{FINAL}/results/losses.json'
losses = []
if os.path.exists(lf):
    with open(lf, 'r') as f:
        losses = json.load(f)
        print(f'Loaded {len(losses)} loss values from {lf}')
else:
    print(f'Error: Loss file not found at {lf}. Please ensure training
was completed and the file exists.')

fig = go.Figure()

if len(losses) > 0:
    fig.add_trace(go.Scatter(
        x=list(range(1, len(losses) + 1)),
```

```

        y=losses,
        mode='lines',
        name='Training Loss',
        line=dict(color='#3498db', width=1.5)
    ))

fig.update_layout(
    title=dict(
        text='Training Loss - Final CSRNet (Plan 1 Model +
A+B+C(India) Fine-tune)',
        font=dict(size=16, weight='bold')
    ),
    xaxis_title='Epoch',
    yaxis_title='MSE Loss',
    legend_title='Legend',
    hovermode='x unified'
)

fig.update_xaxes(showgrid=True, gridwidth=1, gridcolor='LightGrey')
fig.update_yaxes(showgrid=True, gridwidth=1, gridcolor='LightGrey')

fig.show()

Loaded 462 loss values from
/content/drive/MyDrive/DLENDSEM/final_v2/results/losses.json

```